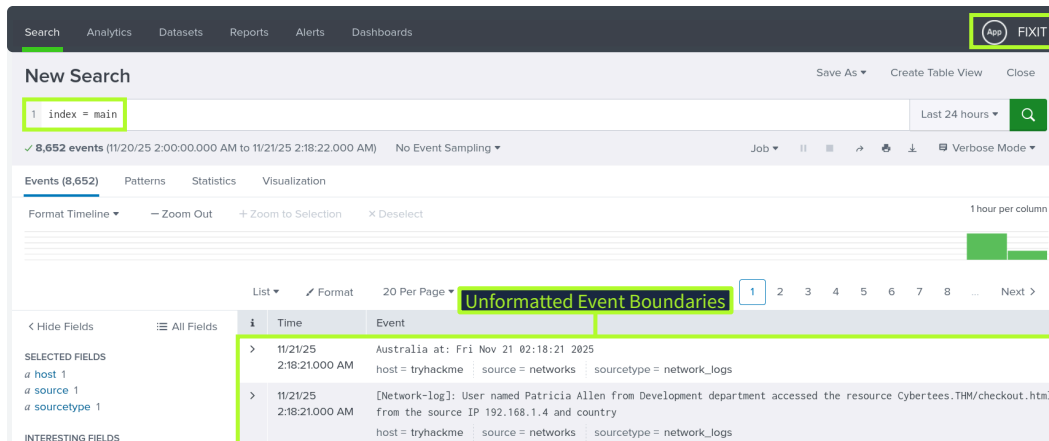


L2-2.5. Splunk: Fixit

In this scenario, you've just completed your third screening interview for a SOC Level 2 role at MSSP Cybertees Ltd, and you're now faced with the final assessment to test your knowledge. You'll be given access to a Splunk instance receiving network logs from an unknown source. The data isn't arriving in a usable state, so before you can analyze what's happening on the network, you must Fixit!

Phase 1: Fixing Event Boundaries

The first phase of your challenge is to fix the event boundaries for the incoming logs. As seen in the screenshot below and in your Splunk instance, the raw data is being ingested and Splunk cannot determine where one event ends and the next begins, making the data impossible to analyze. Go ahead and jump into the Fixit app's configuration files to get started!



Phase 2: Extracting Custom Fields

The next phase of your challenge requires the extraction of meaningful fields from your client's event data. You can accomplish this by updating the Fixit app's configuration files or by creating field extractions directly through the Splunk UI.

Use the sample logs below to help extract the following fields

- Username
- Department
- Domain
- URI
- SourceIP
- Country

[Network-log]: User named Emily Clark from Finance department accessed the resource Cybertees.THM/contact.html from the source IP 192.168.1.4 and country

Japan at: Mon Dec 1 10:13:38 2025

[Network-log]: User named Robert Wilson from HR department accessed the resource Cybertees.THM/signup.html from the source IP 10.0.0.2 and country

Germany at: Mon Dec 1 10:13:42 2025

[Network-log]: User named Patricia Allen from Finance department accessed the resource Cybertees.THM/checkout.html from the source IP 172.16.0.1 and country

Mexico at: Mon Dec 1 10:13:48 2025

Phase 3: Analyzing Event Data

Once the log data is flowing in correctly and the fields have been extracted, it's time to begin your analysis. Using the available data, apply your skills to uncover what's happening on the network!

Answer the questions below

1. What is the full path to the Fixit app directory in your instance? (in the terminal you can find that in the splunk/apps directory)
/opt/splunk/etc/apps/fixit

```
root@tryhackme: /opt/splunk/etc/apps/fixit
File Edit View Search Terminal Help
root@tryhackme:/opt/splunk/etc/apps/fixit# pwd
/opt/splunk/etc/apps/fixit
root@tryhackme:/opt/splunk/etc/apps/fixit#
```

- Investigate the inputs.conf configuration file of the Fixit app. What is the full path of the network-logs script? (the inputs.conf is in the default folder)
/opt/splunk/etc/apps/fixit/bin/network-logs

```
root@tryhackme: /opt/splunk/etc/apps/fixit/default
File Edit View Search Terminal Help
root@tryhackme:/opt/splunk/etc/apps/fixit# pwd
/opt/splunk/etc/apps/fixit
root@tryhackme:/opt/splunk/etc/apps/fixit# ls
bin default local metadata
root@tryhackme:/opt/splunk/etc/apps/fixit# cd default/
root@tryhackme:/opt/splunk/etc/apps/fixit/default# ls
app.conf data inputs.conf
root@tryhackme:/opt/splunk/etc/apps/fixit/default# cat inputs.conf
[script:///opt/splunk/etc/apps/fixit/bin/network-logs]

index = main
source = networks
sourcetype = network_logs
interval = 1
root@tryhackme:/opt/splunk/etc/apps/fixit/default#
```

- Which Splunk stanza setting will you use to define the event boundaries for the scenario logs? BREAK_ONLY_BEFORE

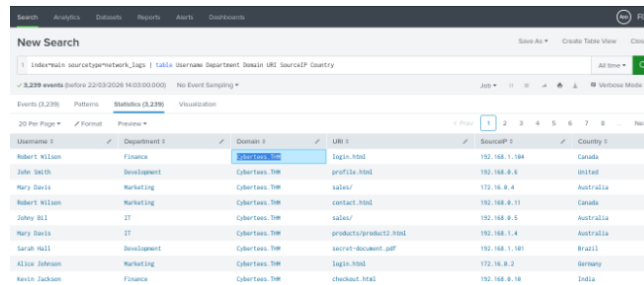
```
root@tryhackme: /opt/splunk/etc/apps/fixit/default
File Edit View Search Terminal Help
root@tryhackme:/opt/splunk/etc/apps/fixit/default# cat ../local/props.conf
[network_logs]
SHOULD_LINEMERGE = true
BREAK_ONLY_BEFORE = ^\[Network-log\]
REPORT-network = network_logs_extract
root@tryhackme:/opt/splunk/etc/apps/fixit/default#
```

- Which regex pattern should be used to define the start of each event? ^\[Network-log\]:

```
root@tryhackme: /opt/splunk/etc/apps/fixit/default
File Edit View Search Terminal Help
root@tryhackme:/opt/splunk/etc/apps/fixit/default# cat ../local/props.conf
[network_logs]
SHOULD_LINEMERGE = true
BREAK_ONLY_BEFORE = ^\[Network-log\]
REPORT-network = network_logs_extract
root@tryhackme:/opt/splunk/etc/apps/fixit/default#
```

5. After you've extracted the relevant fields, what Domain appears in the log data? Cybertees.THM

index=main sourcetype=network_logs | table Username Department Domain URI SourceIP Country

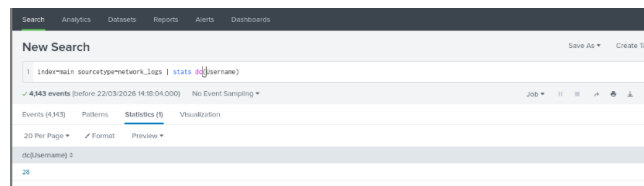


Username	Department	Domain	URI	SourceIP	Country
Robert Wilson	Finance	Cybertees.THM	login.html	192.168.1.184	Canada
John Smith	Development	Cybertees.THM	profile.html	192.168.8.5	United States
Mary Davis	Marketing	Cybertees.THM	sales/	172.16.8.4	Australia
Robert Wilson	Marketing	Cybertees.THM	contact.html	192.168.8.11	Canada
Johny B.1	IT	Cybertees.THM	sales/	192.168.4.5	Australia
Mary Davis	IT	Cybertees.THM	products/product12.html	192.168.1.4	Australia
Sarah Hall	Development	Cybertees.THM	secret-document.pdf	192.168.1.181	Brazil
Alice Johnson	Marketing	Cybertees.THM	login.html	172.16.8.2	Germany
Kevin Jackson	Finance	Cybertees.THM	checkout.html	192.168.8.18	India

6. How many Username field values exist within the events generated? 28

`stats dc(Username)` dc means "distinct count". Counts how many unique usernames exist across all events.

index=main sourcetype=network_logs | stats dc(Username)

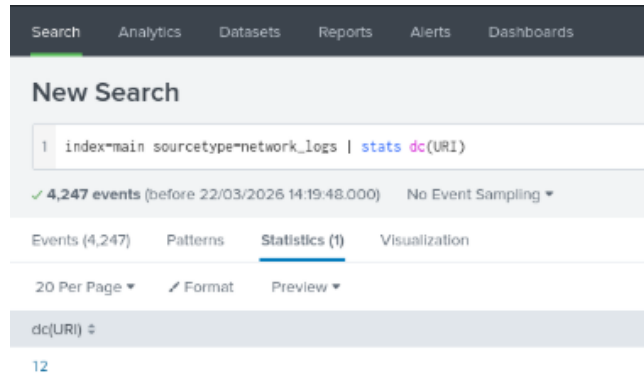


dc(Username)
28

7. How many URI field values were you able to extract from the available logs? 12

`stats dc(URI)` Same idea, counts how many unique URI values (page paths) were extracted.

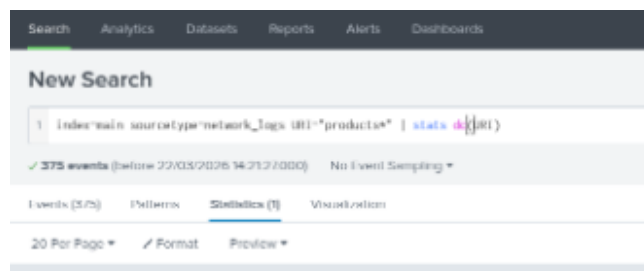
index=main sourcetype=network_logs | stats dc(URI)



8. As you begin analyzing the network traffic, how many individual /products pages appear in the data? 2

`URI="products*" | stats dc(URI)` Filters to only events where the URI starts with "products", then counts the unique ones.

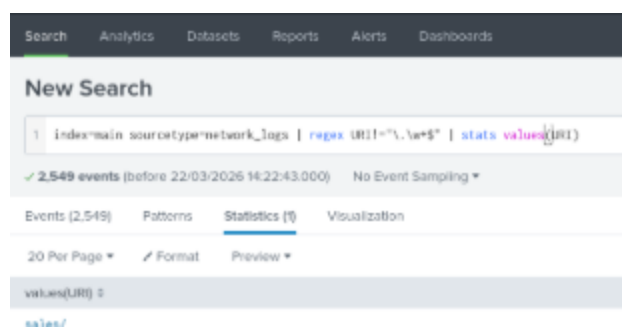
`index=main sourcetype=network_logs URI="products*" | stats dc(URI)`



9. What is the only URI field value found in the event data without a file extension? /sales/

`regex URI!=".\w+$"` Filters out URIs that end with a dot followed by letters (i.e., a file extension like .html, .pdf). What's left is URIs with no extension.

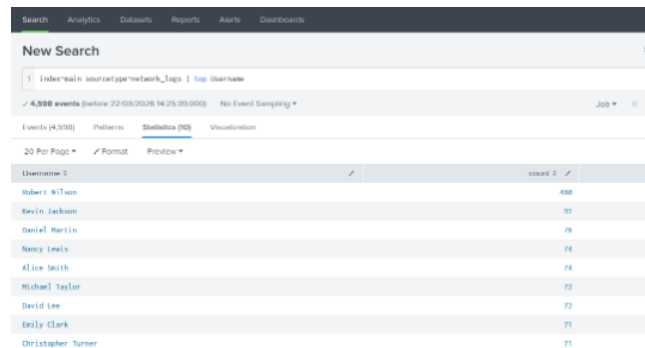
`index=main sourcetype=network_logs | regex URI!=".\w+$" | stats values(URI)`



10. Who is the most active User on the network? Robert Wilson

top Username Returns usernames sorted by how often they appear. The first result is the most active user

```
index=main sourcetype=network_logs | top Username
```



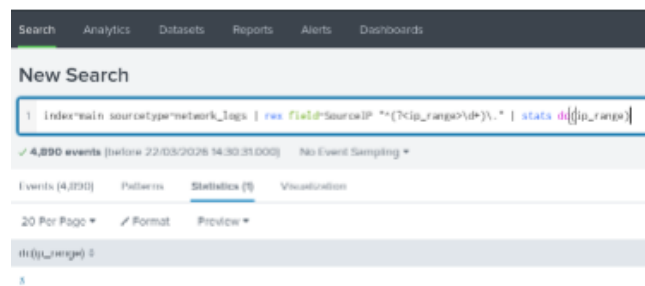
The screenshot shows the Splunk search interface with the query `index=main sourcetype=network_logs | top Username`. The results table lists the top 10 most active users. Robert Wilson is the most active user with 608 events.

Username	count
Robert Wilson	608
Kevin Jackson	35
Daniel Martin	18
Nancy Lewis	14
Alice Smith	14
Michael Taylor	12
David Lee	12
Emily Clark	11
Christopher Turner	11

11. How many unique IP ranges are represented in the observed network traffic?
3

rex field=SourceIP "(?<ip_range>\d+)." Extracts the first octet of the IP (e.g., 10.x.x.x from 192.x.x.x) into a new field called ip_range. Then `dc(ip_range)` counts how many unique ranges there are.

```
index=main sourcetype=network_logs | rex field=SourceIP "(?<ip_range>\d+)." | stats dc(ip_range)
```



The screenshot shows the Splunk search interface with the query `index=main sourcetype=network_logs | rex field=SourceIP "(?<ip_range>\d+)." | stats dc(ip_range)`. The results table shows a single row with the value 3, indicating there are 3 unique IP ranges.

dc(ip_range)
3

12. Which user accessed the secret-document.pdf on your client's server?
Sarah Hall

URI="secret-document.pdf" | stats values(Username) Filters to events where someone accessed secret-document.pdf, then shows which user(s) did it.

```
index=main sourcetype=network_logs URI="secret-document.pdf" | stats values(Username)
```

Search
Analytics
Datasets
Reports
Alerts
Dashboards

New Search

1
index=main sourcetype=network_logs URI="secret-document.pdf" | stats values by username

✓ 70 events (before 22/03/2025 14:28:11.000)
No Event Sampling

Events (70)
Patterns
Statistics (7)
Visualization

20 Per Page
Format
Preview

values[username] 0

Sarah 1611